

B. Sc. (Hons) Semester III Examination 2022 – 23**MATHEMATICS****Paper No. MTB-302 : Differential Equations****Time : 3 hours****Full Marks : 70***(Write your Roll No. at the top immediately on the receipt of this question paper)**The figures in the right-hand margin indicate marks**Answer five questions including Question 1 which is compulsory***1. Answer all parts of this question :**

- (a) Write a non-homogeneous boundary value problem. 2

- (b) Find the value of
- m
- for which
- $y = e^{-2x}(c_1 \cos 2x + c_2 \sin 2x)$
- is a solution of the differential equation

$$m \frac{d^2 y}{dx^2} + 2 \frac{dy}{dx} + 4y = 0$$

2

- (c) Find an integrating factor for the differential equation

$$y(4x + y)dx - 2(x^2 - y)dy = 0$$

2

- (d) Solve the equation

$$(x + a) \frac{dy}{dx} - 3y = (x + a)^5$$

where a is a constant.

2

- (e) Find the singular solution of

$$(xp - y)^2 = p^2 - 1$$

where $p = \frac{dy}{dx}$.

2

- (f) Obtain the solution of

$$\frac{d^4 y}{dx^4} + 8 \frac{d^2 y}{dx^2} + 16y = 0$$

2

- (g) Find

$$\frac{1}{(D - 2)^3} e^{2x} \cos x$$

where $D \equiv \frac{d}{dx}$.

2

- (h) Find the condition for which $y = e^{mx}$ will be a solution of

$$\frac{d^2y}{dx^2} + P(x)\frac{dy}{dx} + Q(x)y = 0$$

2

- (i) Write a second-order differential equation, which has $x = 0$ as regular singular point and $x = 1$ as an irregular singular point. 2
- (j) Prove that $P_n(-x) = (-1)^n P_n(x)$, where $P_n(x)$ are Legendre's polynomials. 2
- (k) For positive integers n , show that—

$$J_{-n}(x) = (-1)^n J_n(x)$$

2

2. (a) Solve the differential equation :

$$y \tan^{-1} \left(\frac{x}{y} \right) dx + \left[y - x \tan^{-1} \left(\frac{x}{y} \right) \right] dy = 0$$

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- (b) Solve the differential equation

$$y = \frac{1}{3}xp + \frac{3x^2}{p}$$

where $p = \frac{dy}{dx}$ completely.

6

3. (a) Solve :

$$\frac{d^3y}{dx^3} - 3\frac{d^2y}{dx^2} + 4\frac{dy}{dx} - 2y = e^x + \cos x$$

6

- (b) Solve :

$$x^3 \frac{d^3y}{dx^3} + 2x^2 \frac{d^2y}{dx^2} + 2y = 10 \left(x + \frac{1}{x} \right)$$

6

4. (a) Solve :

$$\frac{d^2y}{dx^2} - 2 \tan x \frac{dy}{dx} + 5y = e^x \sec x$$

6

- (b) Using the method of variation of parameters, solve the initial value problem :

$$\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + y = \frac{e^{2x}}{(e^x + 1)^2}, \quad y(0) = \log 2, \quad y'(0) = \log 2 + \frac{1}{2}$$

6

5. Solve the equation

$$x^2 \frac{d^2y}{dx^2} + x \frac{dy}{dx} + (x^2 - 9)y = 0$$

in series form about $x = 0$.

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6. (a) Prove that—

$$\int_{-1}^1 [P_n(x)]^2 dx = \frac{2}{2n+1}$$

where $P_n(x)$ are Legendre polynomials.

7

- (b) Prove the following recurrence relation for Legendre's polynomials :

$$(n+1)P_{n+1} = (2n+1)xP_n - nP_{n-1}, \quad n \geq 1$$

5

7. (a) Prove that—

$$xJ'_n = nJ_n - xJ_{n+1}$$

where $J_n(x)$ is Bessel's function.

6

- (b) Without using any recurrence relation, prove that—

$$J^{-3/2}(x) = \sqrt{\frac{2}{\pi x}} \left(\frac{-\cos x}{x} - \sin x \right)$$

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External Portals & Academic Resources:

Examination Portal: <https://bhu-exam.vercel.app/>

Student Resource Center: <https://quashan-me.vercel.app/>

Academic Portfolio: <https://me-aryan.vercel.app>